JESSE COMER

University of Pennsylvania

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RESEARCH

I am broadly interested in ***logic in computer science***, especially the feasibility and limitations of machine reasoning about formal logical properties of computational systems. I use *computational logic* and *formal methods* to specify, analyze, and verify *resilient* and *reliable* systems. My current research focuses on relational program verification for data-centric systems, combining AI-guided search with the reliability of symbolic verification.

EDUCATION

PhD Student, Computer and Information Science, University of Pennsylvania. Aug 2023 - Present

M.S., Computer Science, University of Texas at Austin. Aug 2023

M.Sc., Logic, University of Amsterdam. Aug 2023

B.A., Economics, University of California, Los Angeles. Dec 2016

TEACHING

Mathematical Foundations of Computer Science. Instructor (UPenn). Summer 2024

Algorithms. Teaching Assistant (UT Austin). Spring 2023

Logic, Second Course. Tutor (UCLA). Spring 2016

Logic, First Course. Tutor (UCLA). Fall 2015, Winter 2016

OTHER WORK HISTORY

United States Marine Corps, Reserve. Field Artillery Officer. Jul 2021 - Present

United States Marine Corps. Field Artillery Officer. Jan 2017 - Jul 2021

PREPRINTS

1. **Whiel: A Neurosymbolic Platform for Certified Relational Program Verification**
   J. Comer, F. Shen, S. Zhang, M. Naik, S. Roy, V. Tannen. *arXiv:2611.05612.*

JOURNAL ARTICLES

2. **Verification of Time-Bounded Multiset Rewriting Properties**
   T. Ban Kirigin, J. Comer, M. Kanovich, A. Scedrov, C. Talcott. *JLAMP.* Extended version of WRLA 2024.
3. **Craig Interpolation for Decidable First-Order Fragments**
   B. ten Cate, J. Comer. *LMCS 21(3): 22:1-22:23.* Extended version of FoSSaCS 2024.

CONFERENCE PAPERS

4. **A Unifying Algorithm for Hierarchical Queries**
   M. Abo Khamis, J. Comer, P. Kolaitis, S. Roy, V. Tannen. *PODS 2026.*
5. **The Complexity of Finding Missing Answer Repairs**
   J. Comer, V. Tannen. *ICDT 2026.*
6. **Lovász Theorems for Modal Languages**
   J. Comer. *AiML 2024.*

WORKSHOP PAPERS

7. **Phonological Processes as Modal Transductions**
   T. Yolyan, J. Comer. *SCiL 2026 (ACL 2026 workshop).*

BOOK CHAPTERS

8. **Interpolation in First-Order Logic**
   B. ten Cate, J. Comer. *In B. ten Cate, J. Jung, P. Koopmann, C. Wernhard, and F. Wolter (eds.), Theory and Applications of Craig Interpolation. Ubiquity Press, 2026 (To appear).*

MASTER'S THESIS

9. **Homomorphism Counts, Database Queries, and Modal Logics**
   J. Comer. *ILLC, University of Amsterdam (2023).*
   Committee: M. Gattinger (chair), B. ten Cate (supervisor), N. Bezhanishvili, R. de Haan.

TECHNICAL SKILLS

Lean, Rust, Python; familiarity with automated theorem provers and SMT solvers